

Lottery & Spot Reservation System Specification

Dual Telegram Interface (Bot + Mini App), React Web Admin Dashboard, and Supabase Backend Architecture Plan for Google Antigravity & Engineering Teams.

TARGET PLATFORM

Telegram Bot API & Web

FRONTEND STACK

React (Vite + TS + Tailwind)

BACKEND & DB

Supabase (PostgreSQL + RLS + Storage)

DOCUMENT VERSION

v1.0.0 (Production Blueprint)

1. EXECUTIVE PROJECT SUMMARY

This project specifies an end-to-end lottery and number spot reservation ecosystem. Users interact via a Telegram Bot or a Telegram Mini App (TWA), reserving spot numbers on a visual grid ("Parking Lot" style). Payments are processed through manual transfers (Telebirr, CBE, Bank Transfer), verified by administrators via a web panel. Draws are hosted externally on TikTok Live, while winning announcements and marketing broadcasts are pushed via Telegram.

2. COMPLETE SYSTEM ARCHITECTURE

Client Layer

- **Telegram Bot (Chat UI):** Gatekept onboarding, phone registration, text menu, manual receipt submission, and notifications.
- **Telegram Mini App (React + TWA SDK):** Interactive spot reservation grid, multi-ticket selection cart, product showcase.
- **Admin Web Portal (React Dashboard):** Verification queue, item/round configurator, live broadcast studio, winner declaration engine.

Data & Backend Layer (Supabase)

- **PostgreSQL Database:** ACID-compliant transaction locking for ticket numbers, draw records, and audit logs.
- **Supabase Storage:** Secure storage buckets for uploaded payment receipts and broadcast media (images/videos).
- **Edge Functions / Webhooks:** Telegram webhook handlers, reservation timeout cleanup, and automated item launch broadcasts.

3. USER JOURNEY & DUAL-INTERFACE WORKFLOWS

3.1 Gatekept User Onboarding

/start → Check DB → [Unregistered: Show Only "Share Phone" + "Language"] → Verified → [Registered: Reveal Full Menu]

- **Unregistered User:** The bot displays a persistent keyboard with two options only: [📞 Register (Share Phone Number)] and [🌐 Change Language]. Phone numbers are captured exclusively through Telegram's native contact payload for anti-fraud verification.
- **Registered User:** Once registered, the user profile is stored in Supabase with their selected locale (English, Amharic, Oromo), unlocking the main interactive menu:

[🎟️ Buy Ticket / Open App] | [📋 My Tickets]
[🕒 17 Active Draws & Prizes] | [🌐 Language] | [? Support / Rules]

3.2 Spot Selection: The "Parking Lot" Grid (Telegram Mini App)

- **Visual Number Grid:** Built in React. Displays spots 1 to N with dynamic state colors: **GREEN: AVAILABLE**
AMBER: LOCKED (15M) **BLUE: IN MY CART** **RED/GREY: SOLD / VERIFIED**
- **15-Minute Reservation Lock:** When a user selects spots and clicks "Reserve & Pay", the system transitions those spots to `PENDING_PAYMENT` with a 15-minute expiration timestamp.

- **Multi-Ticket & Repeat Purchases:** Users can buy multiple spots at once or return later to buy more tickets across any active lottery item.

3.3 Manual Payment & Ticket Confirmation

1. The Mini App/Bot displays exact bank/mobile money account details, total price, and a unique Reference ID.
2. User transfers funds and uploads the payment receipt (screenshot / SMS transaction ID) directly in the Telegram chat or Mini App.
3. The receipt and reservation details are pushed to Supabase and immediately appear in the **Admin Review Queue**.
4. Upon Admin approval, spots transition to CONFIRMED, and the bot sends a digital ticket badge to the user. If rejected or expired, spots revert to AVAILABLE.

4. ADMIN PANEL & MARKETING BROADCAST STUDIO

Module	Capabilities	Workflow & Telegram Integration
Payment Verification Queue	• View receipt images & phone numbers. • One-click Approve / Reject with reason. • Timer monitoring for expired locks.	Approval triggers instant bot DM with confirmed ticket numbers. Rejection notifies user with re-upload instructions.
Product & Round Manager	• Create items (e.g. Umbrella, TV, Phone). • Define ticket price, total spots, draw limit. • Set active draw cycles.	Automated Marketing Blast: When a new item is published, bot automatically broadcasts a rich photo card + buy link to ALL registered users.
Manual Broadcast Studio (TikTok & Promos)	• Rich media composer (Photos, Videos, Text). • Manual TikTok Live alert triggers. • Custom CTA button builder.	Admin clicks <i>"Send TikTok Live Alert"</i> → Bot pushes stream link + urgency banner to all active players before the host goes live.
Winner Declaration & Channel Post	• Input winning ticket number. • Automatic user lookup & prize assignment. • Channel broadcast generator.	Bot publishes celebratory winner card with masked phone number (+2519****1234) to public Telegram channel and DMs winner.

5. SUPABASE POSTGRESQL SCHEMA

Deploy the following database schema into Supabase SQL Editor:

```
-- 1. USERS & PROFILES
CREATE TABLE public.users (
  telegram_id BIGINT PRIMARY KEY,
  first_name TEXT,
  last_name TEXT,
  username TEXT,
  phone_number TEXT UNIQUE NOT NULL,
  language_code TEXT DEFAULT 'am', -- 'am', 'en', 'om'
  is_admin BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT timezone('utc'::text, now())
);

-- 2. PRODUCTS & LOTTERY ITEMS
CREATE TABLE public.lottery_items (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title TEXT NOT NULL,
  description TEXT,
  image_url TEXT NOT NULL,
  ticket_price NUMERIC(10, 2) NOT NULL,
  total_spots INT NOT NULL,
  max_draw_cycles INT DEFAULT 1,
  current_cycle INT DEFAULT 1,
  status TEXT DEFAULT 'ACTIVE', -- 'ACTIVE', 'PAUSED', 'COMPLETED'
  created_at TIMESTAMP WITH TIME ZONE DEFAULT timezone('utc'::text, now())
);

-- 3. DRAWS & ROUNDS
CREATE TABLE public.lottery_rounds (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

    item_id UUID REFERENCES public.lottery_items(id) ON DELETE CASCADE,
    round_number INT NOT NULL,
    status TEXT DEFAULT 'OPEN', -- 'OPEN', 'LOCKED', 'DRAWING', 'COMPLETED'
    winning_spot_number INT,
    winner_user_id BIGINT REFERENCES public.users(telegram_id),
    draw_date TIMESTAMP WITH TIME ZONE,
    tiktok_stream_url TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT timezone('utc'::text, now())
);

-- 4. SPOTS & TICKET RESERVATIONS
CREATE TABLE public.tickets (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    round_id UUID REFERENCES public.lottery_rounds(id) ON DELETE CASCADE,
    spot_number INT NOT NULL,
    user_id BIGINT REFERENCES public.users(telegram_id),
    status TEXT DEFAULT 'AVAILABLE', -- 'AVAILABLE', 'PENDING_PAYMENT', 'CONFIRMED', 'CANCELLED'
    reserved_at TIMESTAMP WITH TIME ZONE,
    expires_at TIMESTAMP WITH TIME ZONE,
    receipt_image_url TEXT,
    payment_ref_code TEXT,
    verified_by BIGINT REFERENCES public.users(telegram_id),
    verified_at TIMESTAMP WITH TIME ZONE,
    UNIQUE(round_id, spot_number)
);

-- 5. ADMIN BROADCAST LOGS
CREATE TABLE public.broadcast_logs (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    message_type TEXT NOT NULL, -- 'NEW_ITEM', 'TIKTOK_LIVE', 'CUSTOM_PROMO', 'WINNER'
    text_content TEXT,
    media_url TEXT,
    media_type TEXT, -- 'PHOTO', 'VIDEO', 'NONE'
    button_text TEXT,
    button_url TEXT,
    sent_count INT DEFAULT 0,
    created_by BIGINT REFERENCES public.users(telegram_id),
    sent_at TIMESTAMP WITH TIME ZONE DEFAULT timezone('utc'::text, now())
);

```

6. STEP-BY-STEP IMPLEMENTATION ROADMAP FOR ANTIGRAVITY

Phase 1: Supabase Backend & Database Setup

- Create Supabase Project and execute schema definitions (Users, Items, Rounds, Tickets, Broadcasts).
- Set up Supabase Storage buckets: receipts (Private, accessible by admin) and broadcasts (Public).
- Configure Cron job / Edge function to release expired spot reservations (where status = 'PENDING_PAYMENT' AND expires_at < now()).

Phase 2: Telegram Bot Engine (Node.js / Python)

- Implement /start gatekeeper logic: Query user by telegram_id. If absent, enforce native contact request button.
- Build reply keyboards for language switching, "My Tickets" lookup, and Web App launcher button.
- Implement payment photo receiver handler: stores receipt to Supabase Storage and creates ticket review record.

Phase 3: Telegram Mini App Frontend (React + Vite)

- Integrate @twa-dev / sdk for native Telegram window expansion and haptic feedback.
- Build real-time spot grid component fetching live ticket states from Supabase with polling/realtime subscriptions.
- Implement reservation cart with multi-spot selection and 15-minute lock timer display.

Phase 4: Admin Web Dashboard & Broadcast Studio

- Build Payment Verification page with split-screen view: user details, spots claimed, and high-res receipt zoom.
- Build Lottery Manager: CRUD for items, max draw limits, and round lifecycle controls.
- Build Marketing & TikTok Broadcast Studio: Media uploader (Photos/Videos), custom message editor, and bulk delivery worker.